DESIGN AND ANALYSIS OF ALGORITHMS

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT
1. a)	Define Asymptotic Analysis.	2	1
b)	What are the two key properties that a problem must possess to be solvable by Dynamic Programming?	2	2
c)	Can Topological Sort be applied to a graph with cycles? Justify your answer.	2	
d)	What is an NP-Complete problem?	2	
e)	How does a Randomized Algorithm primarily differ from a deterministic algorithm?	2	
f)	If a problem is NP-hard, does it necessarily mean it is in NP?	2	
2. a)	Solve the recurrence relation $T(n)=7T(n/2)+n^2$ using both the Substitution Method and Master's Theorem.	6	
b)	State and explain all three cases of Master's Theorem used for solving recurrence relations of the form $T(n)=aT(n/b)+f(n)$.	6	
3. a)	Provide a comprehensive comparison of the Greedy and Dynamic Programming algorithmic paradigms.	6	
b)	Apply the Merge and Purge method to solve the following 0/1 Knapsack problem instance: Number of items (n) = 5 Knapsack capacity (m) = 25 Weights (W_i): (W_1, W_2, W_3, W_4, W_5) = (12,8,10,5,15) Profits (P_i): (P_1, P_2, P_3, P_4, P_5) = (15,10,12,6,20)	6	

4. a) Describe Dijkstra's algorithm for finding the single-source shortest paths in a weighted graph with non-negative edge weights. 6

   b) Explain the concept of Topological Sorting for Directed Acyclic Graphs (DAGs). 6

5. a) Explain the proof that the 3-Satisfiability Problem (3-SAT) is NP-Complete. 6

   b) Discuss why NP-complete problems are considered intractable, and describe an approximation approach for solving them. 6

6. a) Show that the absolute approximation problem for set cover problem is NP Hard. 6

   b) Explain the concept of a Polynomial-Time Approximation Scheme (PTAS). 6

7. a) Explain Approximation Algorithms with an example. 6

   b) Write short notes on the following: 6

   i) Branch-and-bound Method

   ii) Time and space trade-offs of an Algorithm.

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